Association Rule Mining

BY

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Rule

$$X \longrightarrow Y$$

- X implies Y
- X is the cause and Y is the effect
- X is antecedent and Y is Consequent
- Y depends on X

The Apriori Algorithm: Basics

- The Apriori Algorithm is an influential algorithm for mining frequent itemsets for boolean association rules.
- Key Concepts :
- •Frequent Itemsets: The sets of item which has minimum support
- •Apriori Property: Any subset of frequent itemset must be frequent.
- •Join Operation: To find L_K , a set of candidate k-itemsets is generated by joining L_{k-1} with itself.

The Apriori Algorithm in a Nutshell

- Find the frequent itemsets: the sets of items that have minimum support
- –A subset of a frequent itemset must also be a frequent itemset

•i.e., if {AB} is a frequent itemset, both {A} and {B} should be a frequent itemset

–Iteratively find frequent itemsets with cardinality from 1 to k (k-itemset)

•Use the frequent itemsets to generate association rules

The Apriori Algorithm : Pseudo code

- Join Step: C_k is generated by joining L_{k-1} with itself
- Prune Step: Any (k-1)-itemset that is not frequent cannot be a subset of a frequent k-itemset
- Pseudo-code:

  C_k: Candidate itemset of size k
  L_k : frequent itemset of size k

```
L1 = {frequent items};
for (k = 1; Lk !=∅; k++) do begin
    Ck+1 = candidates generated from Lk;
    for each transaction t in database do
        increment the count of all candidates in Ck+1
        that are contained in t
    Lk+1 = candidates in Ck+1 with min_support
    end
return ∪k Lk;
```

The Apriori Algorithm: Example

TID	List of Items
T100	I1, I2, I5
T100	I2, I4
T100	I2, I3
T100	I1, I2, I4
T100	I1, I3
T100	I2, I3
T100	I1, I3
T100	I1, I2 ,I3, I5
T100	I1, I2, I3

- Consider a database, D , consisting of 9 transactions.
- Suppose min. support count required is 2 (i.e. min_sup = 2/9 = 22 %)
- Let minimum confidence required is 70%.
- We have to first find out the frequent itemset using Apriori algorithm.
- Then, Association rules will be generated using min. support & min. confidence.

Step 1: Generating 1-itemset Frequent Pattern

Scan D for count of each candidate

Itemset	Sup.Count
{I1}	6
{I2}	7
{I3}	6
{I4}	2
{I5}	2

C_1

Compare candidate support count with minimum support count

Itemset	Sup.Count
{I1}	6
{I2}	7
{I3}	6
{I4}	2
{I5}	2

L_1

- The set of frequent 1-itemsets, L_1 , consists of the candidate 1-itemsets satisfying minimum support.
- In the first iteration of the algorithm, each item is a member of the set of candidate.

Step 2: Generating 2-itemset Frequent Pattern

Generate C_2 candidates from L_1

Itemset
{I1, I2}
{I1, I3}
{I1, I4}
{I1, I5}
{I2, I3}
{I2, I4}
{I2, I5}
{I3, I4}
{I3, I5}
{I4, I5}

C_2

Scan D for count of each candidate

Itemset	Sup. Count
{I1, I2}	4
{I1, I3}	4
{I1, I4}	1
{I1, I5}	2
{I2, I3}	4
{I2, I4}	2
{I2, I5}	2
{I3, I4}	0
{I3, I5}	1
{I4, I5}	0

C_2

Compare candidate support count with minimum support count

Itemset	Sup Count
{I1, I2}	4
{I1, I3}	4
{I1, I5}	2
{I2, I3}	4
{I2, I4}	2
{I2, I5}	2

L_2

Step 2: Generating 2-itemset Frequent Pattern

- To discover the set of frequent 2-itemsets, L_2 , the algorithm uses L_1 *Join* L_1 to generate a candidate set of 2-itemsets, C_2.
- Next, the transactions in D are scanned and the support count for each candidate itemset in C_2 is accumulated (as shown in the middle table).
- The set of frequent 2-itemsets, L_2 , is then determined, consisting of those candidate 2-itemsets in C_2 having minimum support.
- Note: We haven't used Apriori Property yet.

Step 3: Generating 3-itemset Frequent Pattern

Scan D for count of each candidate

Itemset
{I1, I2, I3}
{I1, I2, I5}

C_3

Scan D for count of each candidate

Itemset	Sup. Count
{I1, I2, I3}	2
{I1, I2, I5}	2

C_3

Compare candidate support count with min support count

Itemset	Sup Count
{I1, I2, I3}	2
{I1, I2, I5}	2

L_3

- The generation of the set of candidate 3-itemsets, C_3 , involves use of the Apriori Property.
- In order to find C_3, we compute L_2 *Join* L_2.
- C_3 = L2 *Join* L2 = {{I1, I2, I3}, {I1, I2, I5}, {I1, I3, I5}, {I2, I3, I4}, {I2, I3, I5}, {I2, I4, I5}}.
- Now, Join step is complete and Prune step will be used to reduce the size of C_3. Prune step helps to avoid heavy computation due to large C_k.

Hint:

C3:

{I1, I2, I3} = 2

{I1,I2,I5} =2

{I1,I2,I4} =1

{I1,I3,I5} = 1

{I1,I2,I3,I4} = 0

{I1,I2,I3,I5} = 1

{I1,I2,I4,I5} =0

{I2,I3,I4} = 0

{I2,I3,I5} =1

{I2,I4,I5} = 0

C4:

{I1,I2,I3,I5}=1

L4: Null

Step 3: Generating 3-itemset Frequent Pattern

- Based on the Apriori property that all subsets of a frequent itemset must also be frequent, we can determine that four latter candidates cannot possibly be frequent. How ?
- For example , lets take {I1, I2, I3}. The 2-item subsets of it are {I1, I2}, {I1, I3} & {I2, I3}. Since all 2-item subsets of {I1, I2, I3} are members of L_2, We will keep {I1, I2, I3} in C_3.
- Lets take another example of {I2, I3, I5} which shows how the pruning is performed. The 2-item subsets are {I2, I3}, {I2, I5} & {I3,I5}.
- BUT, {I3, I5} is not a member of L_2 and hence it is not frequent violating Apriori Property. Thus We will have to remove {I2, I3, I5} from C_3.
- Therefore, C_3 = {{I1, I2, I3}, {I1, I2, I5}} after checking for all members of result of Join operation for Pruning.
- Now, the transactions in D are scanned in order to determine L_3, consisting of those candidates 3-itemsets in C_3 having minimum support.

Step 4: Generating 4-itemset Frequent Pattern

- The algorithm uses L_3 *Join* L_3 to generate a candidate set of 4-itemsets, C_4. Although the join results in {{I1, I2, I3, I5}}, this itemset is pruned since its subset {{I2, I3, I5}} is not frequent.
- Thus, $C_4 = \varphi$, and algorithm terminates, having found all of the frequent items. This completes our Apriori Algorithm.
- What's Next ?

  These frequent itemsets will be used to generate strong association rules (where strong association rules satisfy both minimum support & minimum confidence).

Step 5: Generating Association Rules from Frequent Itemsets

- Procedure:
  - For each frequent itemset *"l"*, generate all nonempty subsets of *l*.
  - For every nonempty subset ***s*** of *l*, output the rule "**s → (l-s)**" if **support_count(l) / support_count(s) >= min_conf** where min_conf is minimum confidence threshold.

- Back To Example:

We had L = {{I1}, {I2}, {I3}, {I4}, {I5}, {I1,I2}, {I1,I3}, {I1,I5}, {I2,I3}, {I2,I4}, {I2,I5}, {I1,I2,I3}, {I1,I2,I5}}.

- Lets take *l* = {I1,I2,I5}.
- Its all nonempty subsets are {I1,I2}, {I1,I5}, {I2,I5}, {I1}, {I2}, {I5}.

Hint:

{I1,I2,I3}

{I1},{I2},{I3}, {I1,I3}, {I1,I2}, {I2,I3}, {I1,I2,I3}

{I1} -> {I2,I3}= 2/6=33.3%

{I2}-> {I1,I3} = 2/7=29%

{I3} -> {I1,I2}=2/6=33.3%

{I1,I3}-> {I2}=2/4=50%

{I1,I2}-> {I3}=2/4=50%

{I2,I3}->{I1}=2/4=50%

{I1,I2,I3}->null

Conf: X->Y

Conf= Supp(XUY)/Supp(X)

Hint:

{I1,I2,I5}

{I1},{I2},{I5}, {I1,I5}, {I1,I2}, {I2,I5}, {I1,I2,I5}

{I1} -> {I2,I5}= 2/6=33.3%

{I2}-> {I1,I5} = 2/7=29%

{I5} -> {I1,I2}=2/2=100%

{I1,I5}-> {I2}=2/2=100%

{I1,I2}-> {I5}=2/4=50%

{I2,I5}->{I1}=2/2=100%

{I1,I2,I5}->null

Conf: X->Y

Conf= Supp(XUY)/Supp(X)

Step 5: Generating Association Rules from Frequent Itemsets

- Let minimum confidence threshold is , say 70%.
- The resulting association rules are shown below, each listed with its confidence.
  - R1: I1 ^ I2 → I5
    - Confidence = sc{I1,I2,I5}/sc{I1,I2} = 2/4 = 50%
    - R1 is Rejected.
  - R2: I1 ^ I5 → I2
    - Confidence = sc{I1,I2,I5}/sc{I1,I5} = 2/2 = 100%
    - R2 is Selected.
  - R3: I2 ^ I5 → I1
    - Confidence = sc{I1,I2,I5}/sc{I2,I5} = 2/2 = 100%
    - R3 is Selected.

Step 5: Generating Association Rules from Frequent Itemsets

- R4: I1 → I2 ^ I5
  - Confidence = sc{I1,I2,I5}/sc{I1} = 2/6 = 33%
  - R4 is Rejected.
- R5: I2 → I1 ^ I5
  - Confidence = sc{I1,I2,I5}/{I2} = 2/7 = 29%
  - R5 is Rejected.
- R6: I5 → I1 ^ I2
  - Confidence = sc{I1,I2,I5}/ {I5} = 2/2 = 100%
  - R6 is Selected.

  In this way, We have found three strong association rules.

$$Rule: \; X \Rightarrow Y$$

$$Support = \frac{frq(X,Y)}{N}$$

$$Confidence = \frac{frq(X,Y)}{frq(X)}$$

$$Lift = \frac{Support}{Supp(X) \times Supp(Y)}$$

For an association rule X ==> Y, if the lift is equal to 1, it means that **X and Y are independent**. If the lift is higher than 1, it means that X and Y are positively correlated. If the lift is lower than 1, it means that X and Y are negatively correlated.

A value of lift greater than 1 vouches for high association between {Y} and {X}. More the value of lift, greater are the chances of preference to buy {Y} if the customer has already bought {X}.

Methods to Improve Apriori's Efficiency

- Hash-based itemset counting: A k-itemset whose corresponding hashing bucket count is below the threshold cannot be frequent.
- •Transaction reduction: A transaction that does not contain any frequent k-itemsetis useless in subsequent scans.
- •Partitioning :Any itemset that is potentially frequent in DB must be frequent in at least one of the partitions of DB.
- •Sampling: mining on a subset of given data, lower support threshold + a method to determine the completeness.
- •Dynamic itemset counting: add new candidate itemsets only when all of their subsets are estimated to be frequent

Table 1: A Sample of Marekt Basket Transactions

Transaction ID	Items Bought
1	{a, b, d, e}
2	{b, c, d}
3	{a, b, d, e}
4	{a, c, d, e}
5	{b, c, d, e}
6	{b, d, e}
7	{c, d}
8	{a, b, c}
9	{a, d, e}
10	{b, d}

Support count threshold will be tuned from 2 to Ce[N/2]
N= no. of items;
N=5; 2 to ce[5/2] = ┌2.5┐ =3

Assume Support Count =3

C1	support
{a}	5
{b}	7
{c}	5
{d}	9
{e}	6

L1	support
{a}	5
{b}	7
{c}	5
{d}	9
{e}	6

$C2 = L1 \bowtie L1$

L1	support
{a}	5
{b}	7
{c}	5
{d}	9
{e}	6

C2	support
{a,b}	3
{a,c}	2
{a,d}	4
{a,e}	4
{b,c}	3
{b,d}	6
{b,e}	4
{c,d}	4
{c,e}	2
{d,e}	6

L2	support
{a,b}	3
{a,d}	4
{a,e}	4
{b,c}	3
{b,d}	6
{b,e}	4
{c,d}	4
{d,e}	6

L2	support
{a,b}	3
{a,d}	4
{a,e}	4
{b,c}	3
{b,d}	6
{b,e}	4
{c,d}	4
{d,e}	6

C3	support
{a,b,d}	2
{a,b,e}	2
{a,b,c}	1
{a,b,c,d}	0
{a,b,d,e}	2
{a,d,e}	4
{a,c,d}	1
{a,b,c,e}	0
{a,c,d,e}	1
{b,c,d}	2
{b,c,e}	1
{b,c,d,e}	1
{b,d,e}	4
{c,d,e}	2

L3	support
{a,d,e}	4
{b,d,e}	4

C4	support
{a,b,d,e}	2

L4	Support
NULL	

It will be Continued….